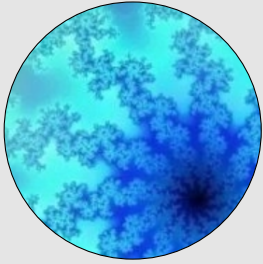


Dr. Alastair Jamieson-Lane

Applied Mathematician and Data Scientist



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About Me

More than a decades experience in teaching, mathematical modelling and programming, as well as collaborating with researchers across industry and academia. Recently finished walking 1300km along the Te Araroa trail from Bluff to Picton. Very sweet tooth.

Areas of Specialization

Mathematical biology, teaching and storytelling, interdisciplinary collaboration.

Interests

Baking, gardening, tramping, storytelling, game design, programming, cake.

Bio

Hey. I'm Alastair. I like riddles and storytelling. I like taking complex intractable problems and breaking them apart into smaller pieces that can be solved using data analysis, or mathematical modelling, or brute force simulation. I enjoy taking abstract, complex concepts and explaining them clearly and concisely, grounding them down to reality. I've collaborated with doctors, social scientists, industry representatives, and government departments, and I hope to do it all again. I can turn understanding into numbers and numbers into understanding.

My contract with Auckland University has recently expired, and now I'm looking for something new to scratch that itch and give me interesting datasets to play with and interesting problems to solve. You want to hire me because I am better at communicating than 90% of analysts, and better at analysis than 90% of communicators.

PROFESSIONAL EXPERIENCE

2020–2025

Professional Teaching Fellow

APPLIED MATHEMATICS · University of Auckland 📍

- Ran courses from first year to graduate level, covering differential equations, numerical methods, probability and programming.
- Mentored a master's student, guiding them through their first exposure to real world research and analysis.
- Re-wrote a 150 page course book designed to clearly communicate with non-experts.
- Produced 100+ interactive presentations aimed at students across a wide range of technical backgrounds.
- Self taught multiple new mathematical techniques in order to present and explain, frequently at short notice.

Some student comments:

- *"I think Alastair has been one of the best lecturers I've had at university. Just overall enthusiastic teaching, and puts in genuine effort."*
- *"Alastair's perky attitude mixed with his excellent sense of humour brought an unbelievable amount of life into all of his lectures. He masterfully balances analogy and technical details in his lectures."*
- *"Good explanations. Slides followed a logical order and are clear to understand"*
- *"His teaching made everything seem doable"*
- *"Alastair has this amazing way of turning this potentially dry maths into a narrative that keeps you engaged and wanting to know what happens next"*
- *"Alastair's lecture slides are an accelerator for pushing the class forward,"*

2019–2020

Postdoctoral Researcher

JUNIOR RESEARCHER · Groningen (Netherlands) and Oldenburg (Germany) 📍

Worked for a research hospital, studying the spread of antibiotic resistance and then later modelling ICU requirements for covid.

- Significant collaboration with a variety of experts and stakeholders outside of mathematics - both learning from their expertise, and focusing in on what aspects of the problem they need clarified and understood.
- Through careful attention to detail, managed to identify a case of fraudulent/dishonest publication in a top scientific journal.
- During the Covid times, I needed to work swiftly and independently, pulling in data from a variety of sources in order to answer key questions in a time critical manner.

WORKSHOPS AND INDUSTRY EXPERIENCE

2024-2026	MINZ - mathematics in industry NZ GROUP CO-LEAD · Christchurch NZ  Week long intensive modelling workshop, working with NZ Police in 2024, Fonterra in 2025 and Malaghan institute in 2026. • Lots of data analysis, getting on top of novel problems and spatio-temporal datasets with a limited time frame. • Constructed novel mathematical models in order to gain insight into real world physical systems. • Extensive collaboration and communication with industry partners. • Professional presentation back to industry partners at the end of the workshop.
2017-18	Data Analysis for BC Safety Authority PYTHON AND NEURAL NETWORKS · Vancouver, Canada  Participated in a week long machine learning sprint, using Tensorflow and python to categorise data provided by the BC-safety authority. Presented results to their data science team back at their office, and then was invited back a <i>second</i> time to present at their annual stakeholder conference.
2017	Programmer CONTRACT WORK · Vancouver, Canada  Coded up a custom data pipeline in R for a colleague in anthropology. Wrote up the user documentation and uploaded to git for public access. Data pipeline is still downloaded and used to this very day.

DEGREES

2019	Applied Mathematics PHD · University of British Columbia 
2014	Mathematics MScI · UBC 
2011	Mathematics BSc · (1st class Hons) GPA 8.26, Canterbury 

PROGRAMMING

R	
Matlab, Julia	
L ^A T _E X	
java, C#	
python	
SQL	

I've learned a dozen programming languages before, and will learn more as needed.

SOME TALKS

2023	How to PhD. One hour talk at auckland graduate seminar introducing, Git, Citation alerts, Overleaf, ArXiv, and how to navigate academic publication
2018	Neural Networks for Identifying Safe (and Unsafe) Systems Invited speaker at an industry/semi-government event
2014	Markov Models of Social Change Guest blog post detailing collaborations with political science.

SELECTED PUBLICATIONS

2022	Comparing optimization criteria in antibiotic allocation protocols. <i>Royal Society Open Science</i>
2019	Timing and Shape of Stochastic Autocatalytic Burst Formation. <i>Chaos</i>
2017	AnthroTools: An R Package for Cross-Cultural Ethnographic Data Analysis. <i>Cross-Cultural Research</i>